10. The integrating factor of the differential equation $\frac{dy}{dx}+P(x)y=Q(x)$ is x, then $P(x)$

(1) x (2) $\frac{x^2}{2}$ (3) $\frac{1}{x}$ (4) $\frac{1}{x^2}$

11. The degree of the differential equation $y(x)=1+\frac{dy}{dx}+\frac{1}{1\cdot 2}\left(\frac{dy}{dx}\right)^2+\frac{1}{1\cdot 2\cdot 3}\left(\frac{dy}{dx}\right)^3+....$ is

(1) 2 (2) 3 (3) 1 (4) 4

12. If p and q are the order and degree of the differential equation $y\frac{dy}{dx}+x^3\left(\frac{d^2y}{dx^2}\right)+xy=\cos x$, when

(1) $p<q$ (2) $p=q$ (3) $p>q$ (4) p exists and q does not exist

13. The solution of the differential equation $\frac{dy}{dx}+\frac{1}{\sqrt{1-x^2}}=0$ is

(1) $y+\sin^{-1}x=c$ (2) $x+\sin^{-1}y=0$ (3) $y^2+2\sin^{-1}x=C$ (4) $x^2+2\sin^{-1}y=0$

14. The solution of the differential equation $\frac{dy}{dx}=2xy$ is

(1) $y=Ce^{x^2}$ (2) $y=2x^2+C$ (3) $y=Ce^{-x^2}+C$ (4) $y=x^2+C$

15. The general solution of the differential equation $\log\left(\frac{dy}{dx}\right)=x+y$ is

(1) $e^x+e^y=C$ (2) $e^x+e^{-y}=C$ (3) $e^{-x}+e^y=C$ (4) $e^{-x}+e^{-y}=C$

16. The solution of $\frac{dy}{dx}=2^{y-x}$ is

(1) $2^x+2^y=C$ (2) $2^x-2^y=C$ (3) $\frac{1}{2^x}-\frac{1}{2^y}=C$ (4) $x+y=C$

17. The solution of the differential equation $\frac{dy}{dx}=\frac{y}{x}+\frac{\phi\left(\frac{y}{x}\right)}{\phi'\left(\frac{y}{x}\right)}$ is

(1) $x\phi\left(\frac{y}{x}\right)=k$ (2) $\phi\left(\frac{y}{x}\right)=kx$ (3) $y\phi\left(\frac{y}{x}\right)=k$ (4) $\phi\left(\frac{y}{x}\right)=ky$

18. If $\sin x$ is the integrating factor of the linear differential equation $\frac{dy}{dx}+Py=Q$, then P is

(1) $\log\sin x$ (2) $\cos x$ (3) $\tan x$ (4) $\cot x$

19. The number of arbitrary constants in the general solutions of order n and $n+1$ are respectively

(1) $n-1,\ n$ (2) $n,\ n+1$ (3) $n+1,\ n+2$ (4) $n+1,\ n$

20. The number of arbitrary constants in the particular solution of a differential equation of third order is

(1) 3 (2) 2 (3) 1 (4) 0

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